Package Drop-Off Log

Week 7 Practice - Problem 5 (Intermediate)

DeliveryNote declares abstract confirmDelivery() and overloads confirmDelivery(signature). ParcelNote and LetterNote directly extend it. logAll() demonstrates upcasting and polymorphic dispatch.

Java solution

```
abstract class DeliveryNote {
    public abstract String confirmDelivery();

    String confirmDelivery(String signature) {
        if (signature == null || signature.trim().isEmpty()) {
            return confirmDelivery();
        }
        return confirmDelivery() + ", signed by " + signature;
    }
}

class ParcelNote extends DeliveryNote {
    private final String trackingId;
    ParcelNote(String trackingId) { this.trackingId = trackingId; }
    public String confirmDelivery() {
        return "Parcel " + trackingId + " delivered";
    }
}

class LetterNote extends DeliveryNote {
    private final String trackingId;
    LetterNote(String trackingId) { this.trackingId = trackingId; }
    public String confirmDelivery() {
        return "Letter " + trackingId + " delivered";
    }
}

public class Main {
    static void logAll(DeliveryNote[] notes) {
        for (DeliveryNote note : notes) {
            System.out.println(note.confirmDelivery());
        }
    }
    public static void main(String[] args) {
        ParcelNote p = new ParcelNote("TRK-1");
        System.out.println(p.confirmDelivery());
        System.out.println(p.confirmDelivery("J. Smith"));
        DeliveryNote ref = p;
        logAll(new DeliveryNote[] { ref, new LetterNote("TRK-2") });
    }
}
```

Sample output

```
Parcel TRK-1 delivered
Parcel TRK-1 delivered, signed by J. Smith
Parcel TRK-1 delivered
Letter TRK-2 delivered
```

Note: The overload calls confirmDelivery() first, then appends the signature so the base delivery message stays in one place.